B.Voc. (SOFTWARE DEVELOPMENT)

Title of the Programme: **Bachelor of Vocation in Software Development leading to degree "B.Voc. (Software Development)**

Software development is the computer programming, documenting, testing, and bug fixing involved in creating and maintaining applications and frameworks involved in a software release life cycle and resulting in a software product. The term refers to a process of writing and maintaining the source code, but in a broader sense of the term it includes all that is involved between the conception of the desired software through to the final manifestation of the software, ideally in a planned and structured process. Therefore, software development may include research, new development, prototyping, modification, reuse, re-engineering, maintenance, or any other activities that result in software products.

Software can be developed for a variety of purposes, the three most common being to meet specific needs of a specific client/business (the case with custom software), to meet a perceived need of some set of potential users (the case with commercial and open source software), or for personal use (e.g. a scientist may write software to automate a mundane task). Embedded software development, that is, the development of embedded software such as used for controlling consumer products, requires the development process to be integrated with the development of the controlled physical product. System software underlies applications and the programming process itself, and is often developed separately. Having understood what software development means we also understand that with the same forms the foundation for the revolution that has taken the world by a storm - the software revolution. In this trillion dollar industry which provides employment to millions of people all over the world, and has potential as well as demand for much more, we intend to provide our students the opportunity to earn and excel.

Software development is a field which has since long, been a favorite. This particular industry is highly rewarding and attracts large number of students since software, its development and maintenance affects us to an ever-increasing extent, both within industry and in our daily lives.
The 3year course block in Software Development gives knowledge and practical skills in the development of software systems of high quality, which is invaluable for software architects, project managers and technical specialists. The demand for knowledgeable experts in software development is steadily increasing, which makes you very competitive nationally as well as internationally, both in industry as well as in academic research.

BACHELOR OF VOCATION

(Software Development)

Semester I								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
GEN-101	Communication Skills	Generic Component	Theory	20	80	-----	-----	6
GEN-102	Fundamentals of Information Technology I	Generic Component	Theory	20	80	-----	-----	6
SD 103	Logic Development Technique	Skill Component	Theory & Practical	10	40	10	40	6
SD 104	Fundamental of Information & Technology II	Skill Component	Theory & Practical	10	40	10	40	6
SD 105	Internet Application	Skill Component	Theory & Practical	10	40	10	40	6
SD 106	Industrial Training & Viva-Voce			--	--	--	--	Non Credits

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

BACHELOR OF VOCATION

(Software Development)

Semester II								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
GEN 107	Soft Skills and Personality Development	Generic Component	Theory	20	80	-----	-----	6
SD 108	Relational Database Management System	Generic Component	Theory	20	80	-----	-----	6
SD 109	RDBMS using MySQL	Skill Component	Theory & Practical	10	40	10	40	6
SD 110	Programming in C Language	Skill Component	Theory & Practical	10	40	10	40	6
SD 111	PC maintenance and trouble shooting	Skill Component	Theory & Practical	10	40	10	40	6
SD 112	Industrial Training & Viva-Voce			--	--	--	--	Non Credits

***Refers to all Generic Components common to all B.Voc. Courses**

Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce

These credits will be evaluated in Semester VI

BACHELOR OF VOCATION

(Software Development)

Semester III								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 301	Value Education And Human Rights	Generic Component	Theory	20	80	-----	-----	6
SD 202	Mathematical Tools for Computer Science.	Generic Component	Theory	20	80	-----	-----	6
SD 203	Object Oriented Programming using C++	Skill Component	Theory & Practical	10	40	10	40	6
SD 204	Data Structure using C++	Skill Component	Theory & Practical	10	40	10	40	6
SD 205	Data communication and networks	Skill Component	Theory & Practical	10	40	10	40	6
SD 206	Industrial Training & Viva-Voce			--	--	--	--	Non Credits

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

BACHELOR OF VOCATION

(Software Development)

Semester IV								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 401	Environmental Studies	Generic Component	Theory	20	80	-----	-----	6
SD 208	Operating System	Generic Component	Theory	20	80	-----	-----	6
SD 209	Web Programming	Skill Component	Theory & Practical	10	40	10	40	6
SD 210	JAVA Programming	Skill Component	Theory & Practical	10	40	10	40	6
SD 211	Computer based accounting	Skill Component	Theory & Practical	10	40	10	40	6
SD 212	Industrial Training & Viva-Voce			--	--	--	--	Non Credits

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

BACHELOR OF VOCATION

(Software Development)

Semester V								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 501	Critical Thinking and Elementary Statistics	Generic Component	Theory	20	80	-----	-----	6
SD 302	Software engineering and quality assurance	Generic Component	Theory	20	80	-----	-----	6
SD 303	Computer graphics and Multimedia Applications	Skill Component	Theory & Practical	10	40	10	40	6
SD 304	ASP.NET using C#	Skill Component	Theory & Practical	10	40	10	40	6
SD 305	Linux operating system	Skill Component	Theory & Practical	10	40	10	40	6
SD 306	Industrial Training & Viva-Voce			--	--	--	--	Non Credits

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

BACHELOR OF VOCATION

(Software Development)

Semester VI								
Paper Code	**Title**	**Generic/Skill Component**	**Theory/ Practical**	**Internal (Theory)**	**External (Theory)**	**Internal (Practical)**	**External (Practical)**	**Credits**
*GEN 601	Entrepreneurship Development Programme	Generic Component	Theory	20	80	-----	-----	6
SD 308	E- Commerce	Generic Component	Theory	20	80	-----	-----	6
SD 309	Emerging Technologies in Computing	Skill Component	Theory & Practical	10	40	10	40	6
SD 310	PHP Programming	Skill Component	Theory & Practical	10	40	10	40	6
SD 311	Network Management	Skill Component	Theory & Practical	10	40	10	40	6
SD 312	Industrial Training & Internship			--	--	--	--	Non Credits

***Refers to all Generic Components common to all B.Voc. Courses**
Industrial Training of 5-6 weeks of 6 credits in each year followed by Report Writing and Viva-voce
These credits will be evaluated in Semester VI

SEMESTER I

Paper - SD103

Logic Development Techniques

Objectives:

1. To develop the logic of the different problems.
2. To equip students with the basic knowledge of logic development techniques.
3. To enhance the skills of students for software development.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

UNIT - I

Logic Development Tools: Algorithm Development: Types of Algorithm, Algorithm of Analysis, Advantage and Disadvantage of Algorithm, Complexity of Algorithm, Big-O Notation.

Flowchart: Types of Flowcharts, Advantage and Disadvantage of Flowchart, Designing Pseudo code, Decision table, Steps for problem solving, Stepwise refinement, Modular Programming: top down & bottom up approach.

Unit II

System Analysis and Design: Definition of problem, Feasibility study, Requirement analysis, Designing, Coding, Testing and Implementation of software project.

UNIT - III

Basics of Programming Language Usage of Character Set, Meaning of Keywords and Identifiers, Role of Data Types, Constants and Variables. Importance of type Casting. Different Types of Operators and their Precedence, Expressions, Conditional Statements

Unit IV

Control Statements: Decision Making statements (if, if-else, else if, nested if), Looping Statements (For, While, do-while), Usage of Exit, Continue, Break and Goto Statement. Introduction of array and functions.

References:

Let us C, Yashvant P Kanetkar, Seventh Edition, BPB Publications, New Delhi.
Programming in ANSI C, E. Balagurusami, Fourth Edition, Tata McGraw Hill
Programming in C, Byron S. Gottfried, Second Edition, McGraw Hills.
The C Programming Language, Kernighan & Richie, Second Edition, PHI Publication
Problem Solving and Programming in C, R. S. Salaria, Second Edition
System Analysis and Design , Elias M awad

SEMESTER-I

Paper - SD104

Fundamentals of Information Technology II

Objectives:

1. To familiarize the students with the Fundamentals of Information Technology and its applications.
2. To use computer systems at operating system level and application level.
3. To assess the implications for the markets and organizational change of advances in information technologies

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

Unit I

Operating Systems- Windows and Unix. Fundamentals of windows, anatomy of windows, Operations on window: Opening, Minimizing, Maximizing, Moving, Resizing, Closing; Windows Explorer, Folders: Creating, deleting, copying, renaming folders, folder properties; Icons, Menu, Taskbar, Control panel, Recycle bin. Overview of UNIX structure, general purpose UNIX commands such as date, echo, cal, bc, pwd, passwd; file and directory commands such as ls,
mkdir, cp, mv, rm. process management commands such as ps, kill, nohup, communication commands such as news, mesg, wall; working with editor introduction to shell programming

Unit II

Word Processing Package: Basics of Word Processing; Word Processing Basics; Text creation, Manipulation, Finding and replacing, Formatting of text; Printing of word document, Page Layout: Margin setting, Alignments, Adding Borders and shading, Adding Headers and Footers, Setting up Multiple columns, Working with tables, Spell check, Auto Correct, Grammar facility, Retrieving often used text; Auto text character formatting, language setting and thesaurus; Macros; Mail merge.

Unit III

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of numbers, text and formulae, Moving data in a worksheet, Moving around in a worksheet, Selecting Data Range. Using the interface (Toolbars, Menus), Editing Basics, Working with workbooks, Cell referencing: Absolute, Relative and Mixed; Formatting and Calculations :using Autofill, Working with Formulae, Efficient Data Display with Data formatting (number formatting, date formatting etc.), Function: different types of functions and their usage Worksheet Printing. Working with Graphs and Charts: Creating Chart and graphs, using chart wizard, sizing and moving parts, updating charts, Changing chart types, Formatting Charts. Database Management; Finding records with Data form, Adding/Deleting Records, Filtering Records in a worksheet, Sorting; Macros: Creating Macros, Record Macros, Running Macros, Assigning Macros to Buttons.

Unit IV

Presentation Packages: Basics, General Features, Creating a presentation, Different types of slide views, Master Slides and its use, Formatting Slides: slide design, Layout and background; Animation effect, Transition effect, timing effects, Macros.

Database Package: Introduction to Database, Tables, Data Types, Attributes, Records; Overview of MS-ACCESS, Creating Database, Creating Tables, Data types, Importing and Exporting data, using Wizards, Creating forms and queries.

References:

Sinha, P.K.: Computer Fundamentals
Rajaraman, V.: Fundamentals of Computers.
Mathur Rajiv: DOS 6.2 Quick Reference, Galgotia.
Texali, R.K.: PC Software made simple, Tata McGraw Hill.

SEMESTER-I

Paper - SD105

Internet Application

Objectives:

1. To analyze the performance of an Internet application protocol and its relation to the underlying network protocols.
2. To promote educational applications of the Internet technology for the benefit of government, colleges, polytechnics, universities and other educational institutions, industry and the public at large**.**

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

UNIT – I

Internet and Intranet: Introduction, History, Services of internet, working of internet, intranet, working of intranet, difference between internet and intranet**.**

E–Mail: Concept, Advantage and disadvantage, structure of an E–Mail message, working of E– Mail (sending and receiving messages), managing E–Mail, and Implementation of outlook express.

UNIT - II

Internet Protocol: Introduction, File Transfer Protocol (FTP), Gopher, Telnet, other protocols Like HTTP and TCP/IP.

WWW: Introduction, working of WWW, Web.

UNIT - III

Search Engine: Introduction to search engine, component of search engine, working of search engine, Web directory, difference between search engine and web directory.

UNIT - IV

News Group: Basic concepts of newsgroup. Online transactions like billing system, electronic commerce.

References:

Internet and its Applications by Ackerman.

Internet – The Complete Reference

Computer Networks (5th Edition) Hardcover by Andrew S. Tanenbaum (Author), David J. Wetherall

Data Communications and Networking by Behrouz A. Forouzan

SEMESTER-I

Paper - SD106

Industrial Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week industrial training (SD 106) and the relevant certificate and project report has to be submitted.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

The report should consist of the following:

Cover page including project title, name of the student, name of department and project guide.

Acknowledgement

Contents with page number

Introduction and background of the application

Objectives

System analysis

System feasibility study

Further scope of the project

Bibliography

Appendices

SEMESTER II

Paper - SD108

Relational Database Management System

Objectives:

1. To provide good understanding of database design, modelling and implementation of database management systems.
2. To educate students with the fundamental concepts of Data Base Design, Data Models and different Data Base Languages.
3. To make access to the data easy for the user.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Unit I

Introduction: Database vs File Oriented Approach, Advantages and Disadvantages of DBMS, Data models: Hierarchical, Network and Relational models.

Unit II

Entity Relationship model, Client-Server Architecture, Data Independence. Concept of Relational Algebra and Calculus, Storage Organization for Relations,

Unit III

Relational Algebra: Operations - union, intersection, difference, Cartesian product, projection, selection, division and relational algebra queries; Relational Calculus: Tuple oriented and domain oriented relational calculus and its operations, Normalizationand its forms.

Unit IV

Database Integrity, Security, Concurrency, Recovery. Backup and Recovery of the database. Distributed databases, its design and structure.

References:

An Introduction to Database System, Galgotia Publications By Desia,B.C
Database System Vol I & II, Narosa Publications By Date, C.J
Database System Concepts, McGraw Hill By Henry F. Korth

SEMESTER II

Paper - SD109

RDBMS using MySQL

Objectives:

1. To introduce principles and foundations of databases, including architecture, design issues, integrity control, query processing and optimization, transactions, and concurrency control.
2. To design and implement database for enterprise application.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

UNIT I

Database Fundamentals: What is a database? What is MySQL? , Introduction to MySQL & MySQL workbench.

Database Design: Introduction To Database Design, Learn Database Normalization with the help of a case study, Learn ER Modeling with a Case Study

UNIT II

MySQL Basics: How To Create A Database & MySQL Data Types, How to use SELECT in MySQL, How to query data using Where clause in MySQL, How to use Insert Into in MySQL, How to Delete & Update data in MySQL

Data Sorting: How sorting is done in MySQL, using ORDER BY, DESC and ASC, How to use Group By in MySQL, How to use Wildcards in MySQL, Using Regular Expressions & Wild Cards in MySQL

UNIT III

Functions: Ultimate guide to Functions used in PHP, How to use Aggregate Function in MySQL, All about Null value & Keyword in MySQL, How to use Auto Increment in MySQL, How to use Alter, Drop & Rename function in MySQL, How to use Limit keyword in MySQL

UNIT IV

Queries: Using Sub-Queries in MySQL, How to use Joins in MySQL, How to use Unions in MySQL, How to use Views in MySQL, How to use Index in MySQL

Minor Application Design: Creating an Application using MySQL

References:

- An Introduction to Database System, Galgotia Publications By Desia,B.C
- Database System Vol I & II, Narosa Publications By Date, C.J
- Database System Concepts, McGraw Hill By Henry F. Korth
- C.J. Date, A. Kannan, S. Swamynathan," An Introduction to Database system", Pearson.
- Dr. Madhulika Jain,Vinita Pillai, Shashi Singh and Satish Jain, "Introduction to Database Management", BPB.
- High Performance MySQL: Optimization, Backups, Replication, and More2nd Edition by Baron Schwartz (Author), Peter Zaitsev (Author),
- MySQL in a Nutshell (In a Nutshell (O'Reilly)) 2nd Edition by Russell J. T. Dyer.

SEMESTER II

Paper - SD110

Programming in C Language

Objectives:

1. To help the students in finding solutions to the various real life problems and converting the solutions into computer program using C.
2. To acquire knowledge about the basic concept of writing a program.
3. To learn and acquire the art of computer programming.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Unit I

Fundamental of 'C': Problem Solving steps, Structure of C Program, I/O statements, assignment statements, C character Set, Variables, Operators and Expressions, Standards and Formatted statements, Keywords, Data Types and Identifiers.

Control Structures: Introduction, Decision making with if – statement, if-else and Nested if, while and do-while, for loop. Jump statements: break, continue, goto, switch Statement.

Unit II

Functions: Function Declaration and Definition, types of functions, Call by value and Call by reference, Recursion, Pre-processor Directives.

Unit III

Preprocessor Directives: Introduction and Use, Macros, Conditional Preprocessors and Header Files.

Arrays: Introduction, Types, Operations on Arrays.

Pointers: Introduction, declaration, initialization, pointers and arrays

UNIT IV

Structure and Union: Declaring of structure, accessing structure members, Array of structure, nested of structure, Unions.

Files: introduction, creating, opening of data files, closing a data file and file handling functions.

References:

Let us C, Yashvant P Kanetkar, Seventh Edition, BPB Publications, New Delhi.

Programming in ANSI C, E. Balagurusami, Fourth Edition, Tata McGraw Hill

Programming in C, Byron S. Gottfried, Second Edition, McGraw Hills.

The C Programming Language, Kernighan & Richie, Second Edition, PHI Publication

Object Oriented Programming, Lafore R, Third Edition, Galgotia Publications

Problem Solving and Programming in C, R. S. Salaria, Second Edition

SEMESTER II

Paper - SD111

PC Maintenance & Troubleshooting

Objectives:

1. To equip the students with the knowledge of hardware maintenance, Software installation and handling troubleshooting.
2. To enhance the basic hardware skills of the students.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit-I

Troubleshooting General PC Problems: Introduction, General Troubleshooting rules, Common Problems & Solutions, Preventive Maintenance.

BIOS: Typical Motherboard BIOS, BIOS Features, BIOS & Boot Sequences, BIOS Shortcoming & Compatible Issues, BIOS Troubleshooting, BIOS Upgrades.

Unit-II

Hard Disk: Introduction, Disk Basics, Disk Performance & Characteristics, Drive, Construction, Drive Testing & troubleshooting.

Motherboard & Buses: Introduction, Motherboard Components, Expansion Slots system Bus Functions & Features. Upgrading & Troubleshooting Motherboard, General Bus Troubleshooting.

Unit-III

Basic Memory Concepts: Introduction, Installing Memories, Upgrade Options & Strategies, Replacing Memories with Higher Capacity. Troubleshooting Memory.

Unit-IV

Printers: Printer Technology, How Printer Works, Attaching Printer, Installing Printer Drivers, Preventive Maintenance, Common Printer Problems & Solution

Error Code: Beep Code, Post Code, Post Reader Card.

References:

Upgrading & Repairing PCs: Muller – Prentice Hall – 10th Edition.

Complete PC Upgrade & Maintenance 32Guide: Mark Minasi–BPB Publishers–15th Edition.

SEMESTER II

Paper - SD112

Industrial Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week industrial training (SD 112) and the relevant certificate and project report has to be submitted.

Job Roles:

Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

The report should consist of the following:

Cover page including project title, name of the student, name of department and project guide.
Acknowledgements
Contents with page number
Introduction and background of the application
Objectives
System analysis
System feasibility study
Further scope of the project
Bibliography
Appendices

SEMESTER-III

Paper – GEN 201

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

SEMESTER-III

Paper – SD202

Mathematical Tools for Computer Science

Objectives:

1. To develop arithmetic, algebraic, geometric, and problem-solving skills.

2. To apply problem solving and logical skills.

3. To make the students familiar with several subfields of mathematics (e.g. numerical analysis, topology, operations research).

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit I

Sets and Relations:- Definition of sets, subsets, complement of a set, universal set, intersection and union of sets, De-Morgan's laws, Cartesian products, Equivalent sets, Countable and uncountable sets, minset, Partitions of sets. **Relations**: Basic definitions, graphs of relations, properties of relations.

Unit II

Matrix:- Introduction, Different kinds, matrix addition and scalar multiplication, multiplication of matrices, transpose etc. Square matrices, inverse and rank of a square matrix, differentiation. Integration.

Unit III

Numerical Methods: Solving simultaneous equations using Gauss elimination, Gauss Jordan Methods, Matrix Inversion method.

Unit IV

Linear Programming:- Graphical and Simplex method, Duality in Linear programming, Transportation problems, Assignment and travelling salesman problem.

References:

- Hiller, F.S&Liberman,G.J: Introduction To Operations Research, 2nd Edition Holden Day Inc London.
- Schaum Series: Theory and Problems Of Essential Computer Mathematics, Mc Graw Hill, New York,3rd Edition.
- Salaria, R.S: Computer Oriented Numerical Methods, Khanna Book Publishing Co (P) Ltd, New Delhi.
- Liu, C. L.: Elements of Discrete Mathematics, McGraw Hill.

SEMESTER-III

Paper – SD203

Object oriented programming using C++

Objectives:

1. To help the students gain a better understanding of OO design and program implementation by using OO language features.
2. To make students learn various concepts of object oriented approach towards problem solving.
3. To introduce different techniques pertaining problem solving skills..

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit I

Concepts of Object Oriented Programming: Introduction to OOP, Difference between OOP and Procedure Oriented Programming, Object, Class, Encapsulation, Data Hiding, Abstraction, Polymorphism, Inheritance. Analysis and design of system using Object Oriented Approach, Structure of a C++ Program and I/O streams.

Unit II

Classes: Class declaration: Data Members, Member Functions, Private and Public Members, Data Hiding and Encapsulation, Arrays within a class.

Objects: Creating objects, Accessing class data members, Accessing member functions, Array of objects, Objects as function arguments: Pass by value, Pass by reference, Pointers to Objects.

Constructors and Destructors: Constructors: Declaration and Definition, Default Constructors, Parameterized Constructor, Copy Constructor and Destructors.

Unit III

Polymorphism: Classification, static and dynamic polymorphism, virtual functions, pure virtual functions.

Function Overloading and Operator Overloading: Function Overloading: Declaration and definition. Operator Overloading: Overloading unary, binary, arithmetic, relational operators, Operator Overloading using friend function.

Unit-IV

Inheritance: Extending classes: Concept of Inheritance, base class, derived class, defining derived classes, visibility modes, private, public and protected

References:

- E. Balaguruswamy : Object Oriented Programming with C++, TMH.
- Bjarne Stroustrup : The C++ Programming Language, Addison Wesley.
- Robert Lafore : OOP in Turbo C++,Galgotia.
- Herbert Schildt : C++ The Complete Reference, TMH.

SEMESTER-III

Paper - SD 204

Data structure using C++

Objectives:

1. To teach students various data structures and explain to them algorithms for performing various operations on these data structures.
2. To use various data structures effectively in application programs.
3. To determine which algorithm or data structure to be used in different scenarios.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

Unit I

Introduction of Data Structure: Definition of Data Structure, types of Data Structure. Operations on data structure.

Arrays: Linear and multi-dimensional arrays and their representation, operations on arrays. Sorting and Searching in array.

Unit II

Stacks & Queues: Definition of stacks Sequential and linked representations, operations on stacks, Queues: Sequential representation of queue, linear queue, circular queue, operations on linear and circular queue, linked representation of a queue and operations on it, priority queue.

Linked List: Linear linked list, operations on linear linked list, doubly linked list, operations on doubly linked list.

Unit III

Trees: Inserting a node, deleting a node, AVL trees, Heap Trees Practical implementation.

Unit IV

Graphs: Traversal of a graph (breadth-first search and depth-first search), and applications of graphs.

References:

- Lipschuitz L. Seymour: Data Structure, Schaum Outline Series, TMH, New Delhi.
- Anshuman Sharma.: Data Structures Using C++, Lakhanpal Publishers.
- Salaria, R. S: Test Your Skills in Data Structures; Khanna Book Publishing Co. (P.) Ltd., New Delhi.
- Sofat Sanjeev: Data Structure with C and C++, Khanna Book Publishing Co.

SEMESTER-III

Paper – SD205

Data Communications and Networks

Objectives:

1. To explain the importance of data communications and the Internet in supporting business communications and daily activities.
2. To analyze the services and features of the various layers of data networks.
3. To explain the role of protocols in networking.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer..

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit I

Introduction:- Data Communication, Network, Data Transmission, Transmission Modes, Network Hardware & Software, Network Topologies, OSI reference model, TCP/IP reference model.

Physical Layer:- Transmission Media, Switching, Multiplexing, Modem..

Unit II

Data Link Layer:- Design Issues, Framing, Error detection & Correction Codes, Flow Control (Stop & Wait, Sliding Window), Error Control, SLIP (Serial Line IP) & PPP(Point to Point).

Unit III

Network Layer:- Design Issues, IPv4 & IPv6 addressing, Routing Algorithms, Congestion control algorithms, Internetworking, Fragmentation, Firewalls.

Unit-IV

Application Layer:- Network Security Threats & Security Services. Cryptography and its algorithms, Message Integrity, Message Authentication, Digital Signatures, HTTP vs HTTPs, Proxy Server, Remote Login and File Transfer.

References:

- Computer Networks, Pearson Education by Andrew S. Tanenbaum
- Data Communication & Networking, Tata McGraw Hill. By Behrouz A. Forouzan.
- Data Communication & Networks, Kalyani Publishers by Charanjeet Singh
- Data Communication & Computer Networks, ABS Publications by Anand Nayyar

SEMESTER-III

Paper – SD206

Industrial Training and Viva Voce

Marks: 100

At the end of the semester, students should compulsorily undergo two week industrial training (SD 206) and the relevant certificate and project report has to be submitted.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

The report should consist of the following:

- Cover page including project title, name of the student, name of department and project guide.
- Acknowledgements
- Contents with page number
- Introduction and background of the application
- Objectives
- System analysis
- System feasibility study
- Further scope of the project
- Bibliography
- Appendices

SEMESTER-IV

Paper –GEN 207

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

SEMESTER-IV

Paper –SD208

Operating System

Objectives:

1. To introduce students with the basic concepts of Operating System, its functions and services.
2. To make students understand the underlying principles, techniques and approaches which constitute a coherent body of knowledge in operating systems.
3. To make students understand how operating system is used as a resource manager.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

UNIT I

Introduction to Operating System:- OS, History & Types of OS, Functions/Operations of OS, Users services/jobs, Segmentation, Virtual memory and Page Replacement algorithms, Scheduling:-Process states, virtual processors, interrupt mechanism, scheduling algorithms- Pre-emptive and Non pre-emptive scheduling; Scheduling Algorithms: FCFS, SJFS, Priority scheduling, Multilevel queue scheduling, Multilevel feedback queue scheduling.

UNIT II

System Deadlock:-System Model, Deadlock Characterization: Necessary conditions, Resource Allocation graph, Method for handling deadlock, Deadlock prevention: Mutual Exclusion, Hold and wait, No Pre-emption, Circular wait, Deadlock Avoidance: Safe state, Resource Allocation graph Algorithm, Banker's Algorithm; Deadlock Detection, Recovery from deadlock.

UNIT III

Memory Management: Hierarchy of memory types, Cache memory: Types: Associative memory, direct mapped, set associative.

Memory Allocation: Address binding, Address Space, Memory Protection, Contiguous and Non- Contiguous allocation, Swapping, Fragmentation; Paging: Protection, Shared pages, Techniques for structuring of page table.

UNIT IV

Segmentation: Segmentation with paging; Virtual Memory: Demand paging; Page replacement Algorithms: FIFO, Optimal, LRU, LFU, MFU, Working set, Thrashing;

References:

- Silberschatz and Peter B.Galvin, "Operating System Concepts" Addison Wesley Publishing Company.
- Dhamdhere, —**Systems Programming & Operating Systems"** Tata McGraw Hill
- GaryNutt, **"Operating Systems Concepts",** Pearson Education Ltd.
- Operating System by Madnick Donovan.

SEMESTER-IV

Paper – SD209

Web Programming

Objectives:

1. To introduce the whole range of web technologies starting from HTML, DHTML, JAVA SCRIPT.
2. To focus on practical aspects of the web technologies.
3. To make the students understand the various steps in designing a creative and dynamic website.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

SECTION – A

Basic terminology: About Web Server; Web Client/Browser, Understanding how a Browser communicates with a Web Server, Website, Webpage, Static Website, Dynamic Website, Internet, Intranet, Extranet, WWW, URL

HTML: Structure of an HTML program, Paragraph Breaks, Line Breaks; Emphasizing Material in a Web Page (Heading Styles, Drawing Lines); Text Styles (Bold, Italics, Underline); Other Text Effects (Centering (Text, Images etc.)

Lists: Unordered List, Ordered Lists, Definition lists Adding Graphics to HTML Documents using the Border, Width, Height, Align, ALT Attributes

Tables: Caption Tag, Width, Border, Cell padding, Cell spacing, BGCOLOR, COLSPAN and ROWSPAN Attributes.

SECTION – B

Linking Documents: Anchor tag, External Document References, Internal Document References and Image Maps

Frames: Introduction to Frames: The <FRAMESET> tag, The <FRAME> tag, Targeting Named Frames

DHTML: Introduction to cascading style sheets (CSS), Style tag, Link tag, Types of CSS: In-Line, Internal, External

Forms: Attributes of Form element, Input element, The Text Element, Password, Button, Submit Button, Reset Button, The Checkbox, Radio, TextArea, Select and Option

SECTION – C

Java Script: Introduction and Features of JavaScript, Writing JavaScript into HTML, tokens, data types, variables, operations, control constructs, strings arrays, functions, core language objects, client side objects, event handling. Applications related to client side form validation.

Other Built-In Objects in JavaScript: The String Object, The Math Object, The Date Object;

SECTION – D

Introduction to Dreamweaver: Understanding Workspace Layout, Managing Websites, Creating a Website, Using Dreamweaver Templates, Adding New WebPages, Text and Page Format, Inserting Tables, Lists, Images, Adding Links.

Web Hosting: Understanding Domain Name & Web Space, Getting a Domain Name & Web Space (Purchase or Free), Uploading the Website to Remote Server, Introduction to Open Source Third party FTP Tools

References:

1	Wanger & Wyke	:	Java Script Unleased, Pearson Education, New Delhi.
2	Bayross, Ivan :	:	HTML, DHTML, Java Script by BPB, Latest reprint
3	Schildt , Herbert	:	The Complete Reference Java 2, TMH, Latest reprint
4	Thomas Powell	:	HTML & CSS: The Complete Reference
5	John Pollock	:	JavaScript, A Beginner's Guide
6	Janine C. Warner	:	Dreamweaver CS5 For Dummies Paperback Edition
7	Joseph Lowery	:	Adobe Dreamweaver CS5 Bible Paperback Edition
8	David Powers	:	The Essential Guide to Dreamweaver CS4

SEMESTER-IV

Paper – SD210

Java Programming

Objectives:

1. To write programs in Java using object-oriented paradigm.
2. To familiarize students with the fundamentals of Java programming, program design and problem-solving.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit I

Introduction to java: Definition, Comparison of java with C++, byte code, java virtual machine, constants ,variables, data types, operators, expressions, control structures, concept of class, creating objects, accessing class members, constructors, method overloading.

Unit II

Packages and interfaces: Defining a package, understanding CLASSPATH, access protection: importing packages, interfaces, defining an interface, implementing interfaces, applying interfaces, variables in interfaces. Exception handling: Definitions, types, try and catch multiple try and catch, built in exceptions.

Unit III

Multithreading: Java thread model, thread priorities, synchronizations, messaging, thread class and runnable interface, main thread: creating a thread, implementing runnable extending thread, creating multiple threads.

Unit IV

I/O applets: I/O Basics: streams, predefined streams, reading console I/P, writing console O/P, print writer class, reading and writing files, applet fundamentals, AWT controls, layout managers and menus, string handling and event handling.

References:

- Phillips LEE and Darmell Rick: Computer Graphics, Second Edition, PHI, New Delhi.
- Daniel Dang,: An Introduction to Java Programming, PHI, New Delhi.
- Balaguruswamy, E.: Programming with Java, A Primer, TMH, New Delhi
  Wanger & Wyke,: Java Script Unleased, Techmedia, New Delhi.

SEMESTER-IV

Paper SD211

COMPUTER BASED ACCOUNTING

Objective:

To equip the students with the basic knowledge of accounting and implement it by using software package TALLY.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

UNIT I

Accounting: concepts and conventions, double entry system of accounting, introduction of basic books of accounts of sole proprietary concern, control accounts for debtors and creditors, closing of books of accounts and preparation of trial balance, Final Accounts: Trading, profit and loss accounts and balance sheet of sole proprietary concern with normal closing entries, Introduction to manufacturing account, final accounts of partnership firms, limited company.

UNIT II

Introduction to computerized financial accounting: coding logic and codes required, master files, Transaction files, Introduction to documents used for data collection, processing of different files, outputs obtained,

UNIT III

Introduction to computerised Inventory control, types of inventory and associated documents, Inventory reports-nature and types, Inventory Control : ABC and Ageing analysis, Methods of Stock validation : LIFO, FIFO, actual bases, Interfacing Inventory with Financial Accounting, Purchasing Sub-Systems, Sales Order processing. Introduction to Computerised Payroll & Invoicing Applications, Exposure to: Structure, Processing and Reports, Interfacing these applications to financial Accounting.

UNIT IV

Use of Accounting package Tally: Introduction to Tally, Groups, Ledgers, Vouchers, Orders, Cost Centres and Categories. Stock. Reports in Tally

References:

- Kellock, J.: Elements of Accounting, Heinemann 1st ed., London.
- Rockley, L.E.,: Finance for the Non-Accountant 2nd. Edition, London: Business Book.
- Levy and Sarnat,: Principle of Financial Management, Prentice Hall International, PHI.
- Arnold,: Financial Accounting, Prentice-Hall International (Paperback Edition).
- Horngren and Sundem: Introduction to Financial Accounting, Prentice-Hall International (Paperback Ediion) N.D: PHI.
- Murthy, U.S: Management Finance, 2nd. Edition, Vakils Fefers & Simons Ltd.
- Van Home, James, C.: Financial Management & Policy, Prentice Inc.
- Pandey, I.M.: Financial Management, Vikas Publication, 6th Rev. ed., N. Delhi.

SEMESTER-IV

Paper – SD212

Industrial Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week industrial training (SD 212) and the relevant certificate and project report has to be submitted.

Job Roles:

Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.

The report should consist of the following:

- Cover page including project title, name of the student, name of department and project guide.
- Acknowledgements
- Contents with page number
- Introduction and background of the application
- Objectives
- System analysis
- System feasibility study
- Further scope of the project
- Bibliography
- Appendices

SEMESTER-V

Paper – GEN301

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

SEMESTER-V

Paper – SD302

Software Engineering and Quality Assurance

Objectives:

1. To make students familiar with all the software development principles, models and designing tools required to develop the software.
2. To make students understand theories, methods, and technologies applied for professional software development.
3. To discuss the concepts of software products and software processes.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

Unit I

Introduction: Characteristics, Software Engineering goals, Software Process Models: Waterfall, Spiral, Prototyping, Fourth Generation Techniques and Software Development Life Cycle.

Software Project Management: Software project management, Project planning and control, cost estimation, project scheduling using PERT and GANTT charts.

Unit II

Software Design: Basic issues in software design, modularity, cohesion, coupling and layering, function-oriented software design: DFD and Structure chart, Object-oriented software development, user interface design.

Unit III

Software Testing: Fundamentals of testing, White-box and black-box testing, Test coverage analysis and test case design techniques, Mutation testing, Static and dynamic analysis, Software Metrics, Types of Software Metrics.

Unit IV

Software Quality Assurance Concepts and Standards: Quality Concepts, Quality Control, Quality Assurance, SQA Activities, Software Reviews, Formal Technical Reviews, Review Guidelines, Software Reliability, Software Safety, Quality Assurance Standards, ISO 9000, ISO 9001:2000, Quality Factors, CMM, TQM, Six Sigma, SPICE, Software Quality Assurance Metrics.

References:

- Roger S. Pressman, “Software Engineering – A Practitioner’s Approach “, Sixth Edition, McGraw Hill
- R.E. Fairley, “Software Engineering Concepts”, Paperback Edition, McGraw Hill.
- Jalota, “An Integrated Approach to Software Engineering”, Third Edition, Narosa Publishing House
- Rajib Mall,” Fundamentals Of Software Engineering”, PHI Learning Pvt. Ltd.

SEMESTER-V

Paper – SD 303

Computer Graphics and Multimedia Applications

Objectives:

1. To study the graphic techniques and algorithms and enable the students develop their creativity.
2. To write programs that demonstrates image processing techniques.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

Unit I

Introduction to Computer Graphics: Overview of Graphics Systems, Display Devices, Hardcopy Devices, Interactive Input Devices- Pointing and positioning devices (cursor, light pen, digitizing tablet, the mouse, track balls).

Unit II

Graphics Packages : Studying the Features and Developing Computer Graphics Using Standard Graphics packages like Auto CAD and Paint Brush.

Unit III

Applications of Multimedia and software packages: Multimedia in the Real World, Training and Education, Image Processing. Studying features and use of Multimedia Image Processing authoring tools like Photoshop and Macromedia Director.

Unit IV

Multimedia Technology: Multimedia in use: Introducing multimedia, uses of multimedia, Technology System Components, Multimedia Platforms, Development Tools, Image, Audio, Video, Storage for multimedia and Communications.

References :

1. Hearn and Backes, 1997 : Computer Graphics, Second Edition, PHI, New Delhi.
2. Kanetkar Yashwant, 2003 : Graphics Under 'C', BPB Publications.
3. Judith Jeffcoate, 2007: Multimedia in Practice, Technology and Applications, PHI.
4. Foley, Vandom, Fenier, Hughes, 1 9 9 6 :Computer Graphics, Principles and Practice, IV Edition in 'C'; Addison Wesley Publishers.

5. Ian R. Sinclair, 1994: Multimedia on the PC (with CDROM), BPB Publications.

6. Hillman, David, 1998:Multimedia Technology and Applications, ITP.

7. Vaughan, Tay, 2008:Multimedia Making it Work, Osborne Publishers.

8. Kelly & Bootle, 1989: Turbo 'C', BPB Publications.

SEMESTER-V

Paper – SD304

ASP.Net using C#

Objectives:

1. To make students familiar with development of console, windows and web applications using C# on .NET platform.
2. To Improve object-oriented programming skills through practice and insights gained by studying a new programming language.
3. To gain a working knowledge of the C Sharp programming language and build object-oriented applications using C Sharp.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

UNIT I

Introduction to ASP.NET: .NET Architecture, ASP.NET Fundamentals Compilation Modes of .Net Architecture, Understanding Client Server Architecture, Introduction To Web Technologies.

UNIT II

Standard Rich Controls: Templates in ASP. NET, Composite Data Bound Controls, Designing Website With Master Pages, Maintaining Application State, Microsoft Application Block, OOPS/COM/DLL/Assembly.

UNIT III

Globalization And Localization: Custom Controls, Web User Controls , Caching Application Pages and Data , Security, Java Scripts, JQUERY, AJAX .NET .

UNIT IV

Windows Mobile Application Development: Application Development Mobile Browsers, Android Applications, LINQ, XML , XSLT, Web Services, WPF(SILVERLIGHT), WCF , Deployment.

Suggested Readings/books:

- Jones, Bradley L.: Sams Teach Yourself C# in 21 Days.
- Balagurusamy, E.: Programming in C#, Tata McGraw-Hill.
- Liberty, J.: Programming C#, O'Reilly.
- Schildt, Herbert: The Complete Reference: C#, Tata McGraw-Hill.
- Robinson: Professional C#, Wrox Press.
- Jason Beres: SAMS Teach Yourself Visual Studio.Net in 21 days.

SEMESTER-V

Paper – SD305

Linux operating System

Objectives:

1. To get familiar with Linux system, its commands, files & directories, system, shell programming,
2. To introduce the student with Unix/Linux kernel programming techniques.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

UNIT I

Introduction to Linux : What is Linux, Linux's History, Minimum System Requirements; Installing Linux : Working with Linux, Floppy-less Installation, Boot and Root Disks, Choosing Text or Graphics Installation, Setting up your Hard Drive, Formatting the Partitions, Setting up the Ethernet, Configuration X, Selecting packages to Install, Using LILO; Partitioning the Hard Disk : Linux Swap Space Partitions, Linux's fdisk, Enabling the Swap Space for Installation, Creating the Linux File-system partition, Using LILO .

UNIT II

Using Linux : Starting and Stopping your Linux System, Linux Shutdown Commands, Login, Passwords, Creating a New Login, Logging Out; Trying out your new Login : Linux Error Messages, Search Paths; The who Command, Commands and Programs, Basic Linux Commands : How Linux Commands Work, Command Options, Other Parameters, Input and Output Redirection, National conventions used to Describe Linux commands, Online help available in Linux, The Linux Man pages, Finding keywords in Man pages, The bash shell help facility; Wildcards : * and ?, Environment Variables, Process and How to Terminate them, The process status Commands : ps, The process termination command : kill, the su command, the grep command.

UNIT III

Using the File System: Files Overview, Common types of files, filenames, Directories an Overview, Parent directories and sub-directories, The root directory, How directories are named, The home directory; File and Directory Permissions : File and Directory ownership, User and ownership, Groups, Changing group ownership, File Permissions, UMASK Setting, Changing File Permission, Changing directory permissions. Bash: What is Shell? How the Shell gets Started, The most common Shells; The Bourne Shell: Command-line Completion, Wildcards, Command History, Aliases, Input Redirection, Output Redirection, Pipelines Shell, Prompts, Job control, Customizing bash, bash commands, bash variables, tcsh : An Introduction to tcsh, Command completion, Wildcards, Command History, Aliases, Input and Output Redirection, Pipelines, Prompts, Job Control; Key Bindings, Correcting Spelling Errors, Pre-commands, Change directory Commands, Monitoring Logins and Logouts, Customizing tcsh, tcsh Command Summary, tcsh variables.

UNIT IV

Shell Programming: Creating and Running Shell Programs, Using variables: Assigning a value to a variable, Accessing the value of a variable, Positional Parameters and other Built-In Shell Variables; The Importance of Quotation Marks : The test Command, The tcsh Equivalent of the test command, Conditional Statements : if Statement , case Statement; Iteration Statements : for Statement, while Statement, until Statement, shift Command, select Statement, repeat Statement, Functions.

Editing and Typesetting : Text Editors vi, The vi Editor, Starting vi, vi modes, Inserting Text, Quitting vi, Moving the Cursor, Deleting Text, Copying and Moving Text, Searching and Replacing Text, Setting Preferences.

Suggested Readings/Books:

- Tim Parker: Linux Unleashed, Techmedia Publishing House.
- Norton, P.: Complete guide to LINUX, Techmedia.
- Komarinski, M.: LINUX System Administration Handbook, AW.
- John Goerzen: Linux Programming Bible, IDG Books, New Delhi.
- Venkateshmurthy, M.G.: Introduction to Unix & Shell Programming, Pearson Education.

SEMESTER-V

Paper – SD306

Industrial Training and Viva Voce

At the end of the semester, students should compulsorily undergo two week industrial training (SD 306) and the relevant certificate and project report has to be submitted.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

The report should consist of the following:

- Cover page including project title, name of the student, name of department and project guide.
- Acknowledgements
- Contents with page number
- Introduction and background of the application
- Objectives
- System analysis
- System feasibility study
- Further scope of the project
- Bibliography
- Appendices

SEMESTER-VI

Paper – GE004E307

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

SEMESTER-VI

Paper – SD308

E-Commerce

Objectives:

1. To integrate the knowledge of foundational functional areas of commerce in order to develop a holistic perspective on the role of IT in organizations.
2. To discuss emerging e-commerce topics.
3. To apply the concepts of Internet security and multimedia in e-business applications.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

Unit-I

Introduction - E-commerce. Advantages, Types and Classification of e-commerce, Building the Web site, Internet, Online transaction, Shopping Card, Software, Risk of e-commerce, email Ids.
Digital Signature: Digital Signature and its Application in E-Commerce, E-Commerce Privacy and Trust.

Unit-II

Enterprise Resource Planning: The Emergence of ERP Systems, Business benefits of ERP,ERP Modules and design alternatives, challenges of ERP implementation, Business Processed engineering, ERP system development process.

Unit-III

Mechanism of making payment through internet: Online payment mechanism; Electronic Payment systems, payment Gateways, Visitors to website, tools for promoting websites; Plastic Money: Debit Card, Credit Card.

Unit-IV

Growth of E-Commerce - Present and potential Data and Measurement issue for E-Commerce.
The infrastructure for E-Commerce, Recent and Near terms, Video Conferencing and Internet. Entertainment Education Health Services, Professional Services, Publishing, Financial Services, Web Site Evaluation.

Suggested Readings/books:

- KalaKota, Ravi & Whinston, Andrew B.: Frontiers of Electronic Commerce; Addison Wesley.
- Mary Sumner: Enterprise Resource Planning, Pearson Education.
- Bajaj, Kamlesh & Nag, Debjani: E-Commerce-The Cutting Edge of Business; TataMcGraw-Hill.
- Rajesh Ray: Enterprise Resource Planning – Text and cases, TMH.
- Minoli, Daniel & Minoli, Emma: Web Commerce Technology Handbook; Tata McGraw Hill.

SEMESTER-VI

Paper – SD309

Emerging Technologies in Computing

Objectives: This course enables students to be familiar with Emerging Technologies such as Parallel Computing, Cloud Computing, Green Computing and Data Mining & data warehousing.

Job Roles:- Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator. C# .net software developer, System Analyst, Software Quality engineer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

All questions carry equal marks.

Unit-I

Parallel Computing: Concept of Parallel Computing, Flynn's Classical Taxonomy: SISD, SIMD, MISD, MIMD; Parallel Computer Memory Architectures: Shared Memory, Distributed Memory, Hybrid Distributed-Shared Memory; TDB and BNP Class of Scheduling Algorithms.

Unit-II

Cloud Computing: Introduction and use, Architecture, Service Models: infrastructure as a service, platform as a service, and software as a service.

Unit-III

Green Computing IT Fundamentals: Business, IT, and the Environment. Green Assets: Buildings, Data Centers, Networks, and Devices

Unit-IV

Data mining & Warehouse: KDD versus data mining, Stages of the Data Mining Process-task premitives, Data Mining Techniques -Data mining knowledge representation – Data mining query languages, Integration of a Data Mining System with a Data Warehouse

Suggested Readings/books:

1.Jiawei Han and Micheline Kamber, Data Mining: Concepts and Techniques, Morgan Kaufmann Publishers, ISBN: 1558604898

2. Alex Berson and Stephen J. Smith, " Data Warehousing, Data Mining & OLAP", TataMc Graw Hill.

3. G. K. Gupta, "Introduction to Data Min Data Mining with Case Studies", Easter Economy Edition, Prentice Hall of India,

4. Joshy Joseph, Craig Fellenstein : Grid Computing, Pearson Education.

5. M. Sasikumar, Dinesh Shikhare, P. Ravi Prakash: Introduction to Parallel Processing,PHI.

6. John Lamb, "The Greening of IT", Pearson Education.

7. Jason Harris, "Green Computing and Green IT- Best Practices on regulations & industry", Lulu.com.

SEMESTER-VI

Paper – SD310

PHP Programming

Time: 3 hrs

Objectives:-This course will enable the students to learn the PHP language for website development.

Job Roles:- Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator. C# .net software developer, System Analyst, Software Quality engineer.

UNIT – I

Introducing PHP – Basic development Concepts – Creating first PHP Scripts – Using Variable and Operators – Storing Data in variable – Understanding Data types – Setting and Checking variables Data types – Using Constants – Manipulating Variables with Operators.

UNIT – II

Controlling Program Flow: Writing Simple Conditional Statements - Writing More Complex Conditional Statements – Repeating Action with Loops – Working with String and Numeric Functions.

UNIT – III

Working with Arrays: Storing Data in Arrays – Processing Arrays with Loops and Iterations – Using Arrays with Forms - Working with Array Functions – Working with Dates and Times.

UNIT – IV

Using Functions and Classes: Creating User-Defined Functions - Creating Classes – Using Advanced OOP Concepts. Working with Files and Directories: Reading Files-Writing Files-Processing Directories. Working with Database and SQL : Introducing Database and SQL- Using MySQL-Adding and modifying Data-Handling Errors – Using SQLite Extension and PDO Extension. Introduction XML—Simple XML and DOM Extension.

SUGGESTED READINGS

Christopher J.Goddard, Mark White, —Mastering VB Script□, Galgotia publications, New Delhi.
Lee Purcell, Mary Jane Mara, —The ABCs of Javascript
Steven Holzner, —PHP: The Complete Reference

Note: Latest and additional good books may be suggested and added from time to time.

SEMESTER-VI

Paper – SD311

Network Management

Objectives:-This course will enable the students to manage the network devices and configuring servers

Job Roles:- Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator. C# .net software developer, System Analyst, Software Quality engineer.

Instructions:

The syllabus of this paper has been divided into FOUR units.

Examiner will set a total of **NINE** questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.

Unit-I

Management of Network Devices:- Managing network devices :- Routers, Switches, hubs Startup and Configuration

Unit-II

Configuring and installing Window based servers:- Configuring IP addresses , Installing and Configuring Windows Server - Preparing for Installation, Creating windows server boot disk, Installing windows server, Configuring server/ client

Unit- III

Configuring & installing Linux based Servers:- Configuring IP addresses , steps for installing linux based server, installation of drivers , Configuring server on the network

Unit-IV

Windows Networking: Sharing files and resources:- User Accounts, Groups, Permissions & Their Role in Sharing, The Basic Concepts in Network Sharing, Customizing Your Network Sharing Settings, How to Share Devices With Others On the Network.

References:-

1. **Network Management:** Concepts and Tools:- Edited by ARPEGE Group: CHAPMAN & HALL
2. **Installing & Configuring Windows Server 2012 R2:-**Craig Zacker
3. **Complete Linux Servers:-Installation & Configuration:-** Chetan Soni

SEMESTER-VI

Paper – SD312

Industrial Training and internship

The students will have to undergo 8 week training and prepare a project during internship. They shall submit the project report at the completion of the internship. The project will involve development of application/system software in industrial/ commercial/ scientific environment. The objective of this course is to translate a problem by designing software.

Job Roles:

Java Programmer Trainee, Linux Administrator, Web Programmer Trainee, Database administrator, Network Administrator, C#.net software developer, System Analyst, Software Quality engineer

The report should consist of the following:

- Cover page including project title, name of the student, name of department and project guide.
- Acknowledgements
- Contents with page number
- Introduction and background of the application
- Objectives
- System analysis
- System feasibility study
- Further scope of the project
- Bibliography
- Appendices